Total No. of Questions : 8 | | Total No. of Printed Pages : 3

Paper Code : 21308

F-408

B.C.A. (Second Semester)

Examination, 2023

(New Course)

Paper No. BCA-N-203

MATHEMATICS-II

Time : Three Hours] *[Maximum Marks : 70*

Note : Attempt any **five** questions. **All** questions carry equal marks. Symbols used are as usual.

1. (a) Show that every infinite bounded set has at least one limitting point.

   (b) The intersection of a finite collection of open sets in R is an open set in R.

2. (a) Show that $\lim_{(x,y)\to(-1,2)} \frac{x^3+y^3}{x^2+y^2} = \frac{7}{5}$

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   (b) Find the maximum value of $(ax + by + cz)e^{-\alpha^2x^2-\beta^2y^2-\gamma^2z^2}$

3. (a) Prove that $\lim_{n\to\infty} \frac{1}{n}[(n+1)(n+2)\ldots(2n)]^{1/n} = \frac{4}{e}$

   (b) Test for the convergence or divergence of the series Σu_n where $u_n = \sqrt{(n^4+1)} - \sqrt{(n^4-1)}$

4. (a) Test for convergence the series

   $x \log x + x^2 \log(2x) + x^3 \log(3x) + \ldots\ldots\ldots + x^n \log(nx) + \ldots\ldots$

   (b) Show that a monotonic increasing sequence is convergent iff it is bounded.

5. (a) Expand log sec x by Maclaurin's theorem.

   (b) If $\phi(x) = \sqrt{x}, f(x) = \frac{1}{\sqrt{x}}$, then show that c is the geometric mean of a, b in Cauchy's M.V.T. while c is the harmonic mean of a, b if $\phi(x) = \frac{1}{x^2}, f(x) = \frac{1}{x}$

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6. State and prove comparison test.

7. (a) Show that the function

$$f(x) = \begin{cases} \dfrac{xe^{1/x}}{(1+e^{1/x})} & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$ is continuous

at x=0 but not differentiable at x=0.

(b) Evaluate $\lim_{x \to 0} \left(\frac{\tan x}{x}\right)^{1/x}$

8. (a) Show that if the perimeter of a triangle is constant, its area is a maximum when it is equilateral.

(b) Show that the sequence $\left\langle \frac{2^n}{n!} \right\rangle$ is a monotonic decreasing sequence. Also show that it is bounded.

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MATHEMATICS-II

Time : Three Hours] *[Maximum Marks : 70*

Note : Attempt any **five** questions. **All** questions carry equal marks. Symbols used are as usual.

1. (a) Prove that the system Q of rational numbers has the Archimedean property i.e. is an Archimedean ordered field.

   (b) Show that any open interval is a neighbourhood of each of its points.

2. (a) Find the limit points of the set

$$S = \left\{ \frac{n}{n+1} ; n \in \mathbb{N} \right\}$$

   (b) Prove that the intersection of an arbitrary collection of closed sets is closed.

3. (a) Evaluate : $\lim_{x \to 0} \frac{2^x - 1}{(1+x)^{1/2} - 1}$

   (b) Examine the following function for continuity at x=0 and x=1.

$$f(x) = \begin{cases} x^2 ; & x \leq 0 \\ 1 ; & 0 < x \leq 1 \\ \frac{1}{x} ; & x > 1 \end{cases}$$

4. (a) If $\langle s_n \rangle$ is a Cauchy sequence of real numbers which has a subsequence converging to ℓ, prove that $\langle s_n \rangle$ itself converges to ℓ. https://www.mjpruonline.com

   (b) Prove that the set of limit points of every sequence is a closed set.

5. (a) Expand tan x by Maclaurin's theorem.

   (b) Verify Cauchy's Mean Value Theorem for the functions x^2 and x^3 in the interval [1, 2].

6. (a) Let $f:[0, 1]\to R$ be defined by

$f(x)= (x-1)^2+2$, $\forall x\in[0, 1]$.

Find the equation of the tangent to the graph of this curve which is parallel to the chord joining the points (0,3) and (1,2) of the curve.

(b) Test for convergence of the series whose n^{th} term is given by $\sqrt{n^2+1}+\sqrt{n^2-1}$

7. (a) Evaluate : $\lim_{x\to 0}\left[\frac{(1+x)^{1/x}-e}{x}\right]$

(b) Show that if the perimeter of a triangle is constant, its area is maximum when it is equilateral.

8. (a) Find the maxima and minima of $(4-3x)^2e^x$.

(b) Show that the sequence $\langle s_n\rangle$ where $s_n=\sin n\pi\theta$ and θ is a rational number such that $0<\theta<1$, is not convergent.

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Paper Code : 21308

F–408

B. C. A. (Second Semester) EXAMINATION, 2019

(New Course)

Paper No. BCA—N—203

MATHEMATICS—II

Time : Three Hours] [*Maximum Marks : 70*

Note : Attempt any *five* questions. All questions carry equal marks. Symbols used are as usual.

1. (a) Prove that a bounded set has greatest and least limit points.

   (b) Prove that the derived set of every set is a closed set.

2. (a) If :

$$f(x) = \begin{cases} x+2 & \text{when} \quad n < 1 \\ 4x-1 & \text{when} \quad 1 \le n \le 3 \\ x^2+5 & \text{when} \quad n > 3 \end{cases}$$

find $\lim_{n \to 1} f(x)$ and $\lim_{n \to 3} f(x)$ if they exist.

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   (b) Evaluate :

$$\lim_{x \to 0} \left[\frac{(1+x)^{1/x} - e}{x} \right].$$

3. (a) Check the continuity for $x = 0$ of f defined by :

$$f(x) = \frac{1}{1 - e^{1/x}}$$

when $x \neq 0$ and $f(0) = 0$.

   (b) Show that the sum, difference and product of two functions continuous at a point $x = a$ are continuous at this point.

4. (a) Test for convergence the series whose n^{th} term is :

$$\sqrt{n^3+1} + \sqrt{n^3-1}$$

   (b) Test for convergence and divergence of the series :

$$1 + \frac{3}{7}x + \frac{3.6}{7.10}x^2 + \frac{3.6.9}{7.10.13}x^3 + \frac{3.6.9.12}{7.10.13.16}x^4 \ldots$$

5. State and prove p-series test.

6. (a) Expand $\sin x$ by Maclaurin's theorem.

   (b) If the mean value theorem is :

$$f(b) - f(a) = f'(x_1)(b-a)$$

find x_1 when $f(x) = x^3 - 3x - 1$, $a = -11/7$, $b = 13/7$.

7. (a) Find the maximum and minimum value of $(1-x)^2 e^x$.

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(b) Show that the right circular cylinder of given surface (including the ends) and maximum volume is such that its height is equal to the diameter of the base.

8. (a) Show that if a sequence converges, its limit is unique.

(b) Check the roundedness and convergence of the sequence $z_1, z_2, z_3, \ldots, z_n, \ldots\ldots\ldots$ if $z_n = n \sin \frac{1}{2} n\pi$.

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Paper Code : 21308

F–408

B. C. A. (Second Semester) EXAMINATION, 2018

(New Course)

Paper No. BCA—N–203

MATHEMATICS—II

Time : Three Hours] *[Maximum Marks : 70*

Note : Attempt any *five* questions. All questions carry equal marks.

1. (a) Find the limiting points of the set :

$$\left\{1, -1, 1\frac{1}{2}, -1\frac{1}{2}, 1\frac{1}{3}, -1\frac{1}{3}, \ldots\right\}$$

(b) State and prove Bolzano-Weierstrass theorem.

2. (a) Evaluate $\lim_{x \to 0} \dfrac{(1+x)^{1/3} - (1-x)^{1/3}}{x}$.

(b) Show that a subset A of R is closed iff $\bar{A} = A$.

3. (a) Examine the continuity of the function at $x = 0$.

$f(x) = \dfrac{e^{x^{1/x}}}{1 + e^{1/x}}$, when $x \neq 0$, $f(0) = 0$.

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(b) Show that the sequence $\left\langle \dfrac{n}{n+1} \right\rangle$ is bounded $\forall\, n \in \mathrm{N}$.

4. (a) If $\langle x_n \rangle$ is a sequence such that $x_n > 0\ \forall\, n \in \mathrm{N}$ and $\lim_{n \to \infty} x_n = a$, prove that :

$$\lim_{n \to \infty} (x_1 x_2 x_3 \ldots\ldots x_n)^{1/n} = a$$

(b) Prove that the sequence $\langle x_n \rangle$, where $x_n = \sqrt{n+1} - \sqrt{n}$ converges and find its limit.

5. (a) Test the convergence of the series whose n^{th} term is $\left(\dfrac{1}{n} - \log \dfrac{n+1}{n}\right)$.

(b) Test the series :

$$1 + \frac{3}{2}x + \frac{5}{9}x^2 + \frac{7}{28}x^3 + \cdots\cdots + \frac{2n+1}{n^3+1}x^n + \ldots$$

6. Test the convergence of the series

$$1 + \frac{3}{7}x + \frac{3.6}{7.10}x^2 + \frac{3.6.9}{7.10.13}x^3 + \frac{3.6.9.12}{7.10.13.16}x^4 + \ldots$$

7. (a) State and prove Lagrange's mean value theorem.

(b) Find the expansion of $\log(1+x)$ by using Maclaurin's theorem.

8. (a) Discuss the applicability of Rolle's theorem in the interval $[-1, 1]$ to the function $f(x) = |x|$.

(b) Find the values of a and b in order that

$$\lim_{x \to 0} \frac{x(1 + a\cos x) - b\sin x}{x^3} = 1.$$

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